

SHETH LUJ COLLEGE

Association Rule Mining

- 2 Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- 2 Generate association rules from the frequent itemsets and calculate their support and confidence.
- 2 Interpret and analyze the discovered association rules.

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IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/RULE MINIG.py =====
Transaction Dataset:
Bread Butter Jam Milk
0 True True False True
1 True True False False
2 True False False True
3 True True False True
4 True True True False
5 True False True True
6 False True False True
7 True True False False
8 True True False True
9 True False True False

Frequent Itemsets:
support  itemsets
0 0.9 (Bread)
1 0.7 (Butter)
2 0.3 (Jam)
3 0.6 (Milk)
4 0.6 (Butter, Bread)
5 0.3 (Jam, Bread)
6 0.5 (Milk, Bread)
7 0.4 (Milk, Butter)
8 0.3 (Milk, Butter, Bread)

Association Rules:
```

```
IDLE Shell 3.13.7
3 True True False True
4 True True True False
5 True False True True
6 False True False True
7 True True False False
8 True True False True
9 True False True False

Frequent Itemsets:
support  itemsets
0 0.9 (Bread)
1 0.7 (Butter)
2 0.3 (Jam)
3 0.6 (Milk)
4 0.6 (Butter, Bread)
5 0.3 (Jam, Bread)
6 0.5 (Milk, Bread)
7 0.4 (Milk, Butter)
8 0.3 (Milk, Butter, Bread)

Association Rules:
antecedents consequents support confidence lift
0 (Bread) (Butter) 0.6 0.666667 0.952381
1 (Butter) (Bread) 0.6 0.857143 0.952381
2 (Jam) (Bread) 0.3 1.000000 1.111111
3 (Milk) (Bread) 0.5 0.833333 0.925926
4 (Milk) (Butter) 0.4 0.666667 0.952381
5 (Milk, Bread) (Butter) 0.3 0.600000 0.857143
6 (Milk, Butter) (Bread) 0.3 0.750000 0.833333

>>>
```